

2.7 Cell Integrity

▼ What is the difference between hypoxia and anoxia?

- **Hypoxia is diminished oxygen levels**
- **Anoxia is total oxygen lack**

▼ What is substrate-level phosphorylation?

- It is the production of ATP by the transfer of a high-energy phosphate group from one intermediate substrate to ADP

▼ Give an example of substrate-level phosphorylation in glycolysis.

- Step 7
- 1,3-bisphosphoglycerate $\rightarrow$ 3-phosphoglycerate
- $\text{ADP} + \text{P}_i \rightarrow \text{ATP}$

▼ Where in the mitochondria does the TCA Cycle occur?

- **Mitochondrial matrix**

▼ Where in the mitochondria does oxidative phosphorylation occur?

- In the **inner mitochondrial membrane**.
- **Numerous folds within the cristae** increase the surface area where it can take place.

▼ Why does the reoxidation of FADH₂ produce 2 molecules of ATP but NADH produces 3?

- The electron transport chain has complexes/membrane proteins which are used to pump protons into the intermembrane space to generate ATP via the action of ATP synthase
- **FADH₂ electrons enter at complex II** in the electron transport chain, and bypass complex I, one of the complexes that pump aforementioned protons
- This means that **less ATP is synthesised via the reoxidation of FADH₂** as NADH electrons instead enter the chain at complex I as **NADH results in more H⁺ pumped into the intermembrane space**

▼ What is ubiquinone?

- An **alternative name given to coenzyme Q** (the key coenzyme required for procession of the electron transport chain)

▼ What is succinate dehydrogenase?

- It is an **enzyme** that is found in the inner mitochondrial membrane **that makes up complex II of the electron transport chain.**
- NOTE: Each complex is actually an enzyme

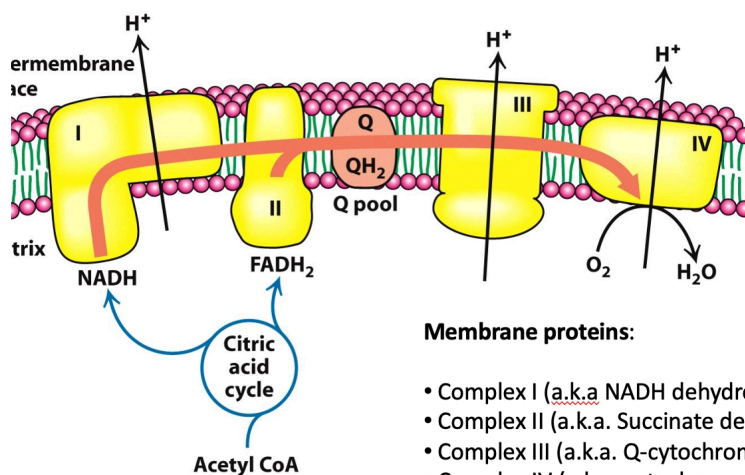
▼ How does succinate dehydrogenase perform its function?

- It **passes electrons directly to coenzyme Q** (ubiquinone) via the action of cofactor FADH₂
- In the process, it also **passes protons to coenzyme Q to form QH₂**

▼ Diagram showing the electron transport chain.

- H⁺ from the cofactors enters the electron transport chain and is pumped out into the intermembrane space by complexes I, III and IV.

The electron transport chain (ETC)

**Mobile Carriers:**

- Co-enzyme Q (a.k.a. ubiquinone)
- Cytochrome C (not shown)

Membrane proteins:

- Complex I (a.k.a. NADH dehydrogenase)
- Complex II (a.k.a. Succinate dehydrogenase)
- Complex III (a.k.a. Q-cytochrome c oxidoreductase)
- Complex IV (a.k.a. cytochrome c oxidase)

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W. H. Freeman and Company

▼ What is a redox couple?

- A substrate that can exist in both oxidised and reduced forms (i.e. NAD⁺/NADH, FAD/FADH₂)

▼ What is reduction potential?

- The ability of a redox couple to accept or donate electrons.

▼ What does a negative/positive redox potential mean?

- **Negative** means the couple is **more likely to donate electrons** and has a greater reducing power than hydrogen
- **Positive** means the couple is **more likely to accept electrons** and has a greater oxidising power than hydrogen

▼ What happens to electrons as they are pumped across the electron transport chain?

- They lose energy

▼ Where does the energy go?

- To be **used to pump protons to the intermembrane space** and form a concentration gradient across the inner membrane
- This allows protons to move across the membrane, and **energy from this movement is transferred to the F1 subunit of ATP synthase to make ATP**

▼ Explanation of energy transfers occurring during oxidative phosphorylation.

- Electrons move **across electron carrier proteins (complexes)** - this movement generates energy
- **Energy is transferred to pump protons** to the intermembrane space
- **Protons move down** the electrochemical/concentration **gradient**, this generates energy
- This **energy is transferred to the F1 subunit of ATP synthase to be used in ATP production.**

▼ Where is ATP synthase located?

- On the inside of the inner mitochondrial membrane

▼ What are the two portions that make up ATP synthase called?

- F₀ (membrane bound) and F₁ (projecting into matrix).

ATP synthase - a molecular turbine powered by protons

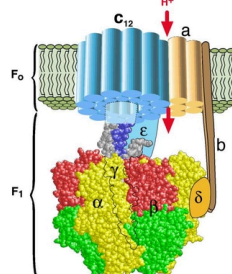
Consisting of:

- F₀ membrane bound
- F₁ projecting into the matrix space.

The F₁ and F₀ parts each consists of three different subunits.

F₀ = a, b and c.

F₁ = α, β and γ



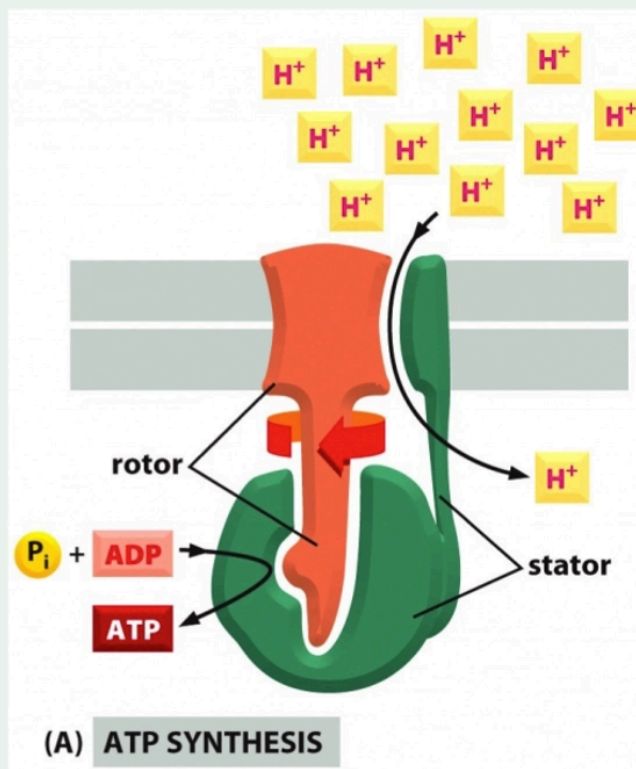
H. Wang and G. Oster (1998). Nature 396:279-282.

▼ What smaller subunits make up these two portions respectively?

- **F₀ - a, b and c**
- **F₁ - alpha, beta and gamma**

▼ What happens to ATP synthase when protons move into the mitochondrial matrix?

- The **F1 subunit starts to rotate**.
 - This is because the alpha, beta and gamma subunits have varying affinities for ADP, Pi and ATP subunits
- **Conformational energy** then flows from the **rotation of this catalytic subunit (F1)** and is **transferred to ADP and Pi, allowing them to make ATP**
- The energy is ultimately converted to chemical energy which remains present in phosphoanhydride bonds of ATP.
- This is **chemiosmosis**



▼ How does ATP synthase decide whether to make or consume ATP?

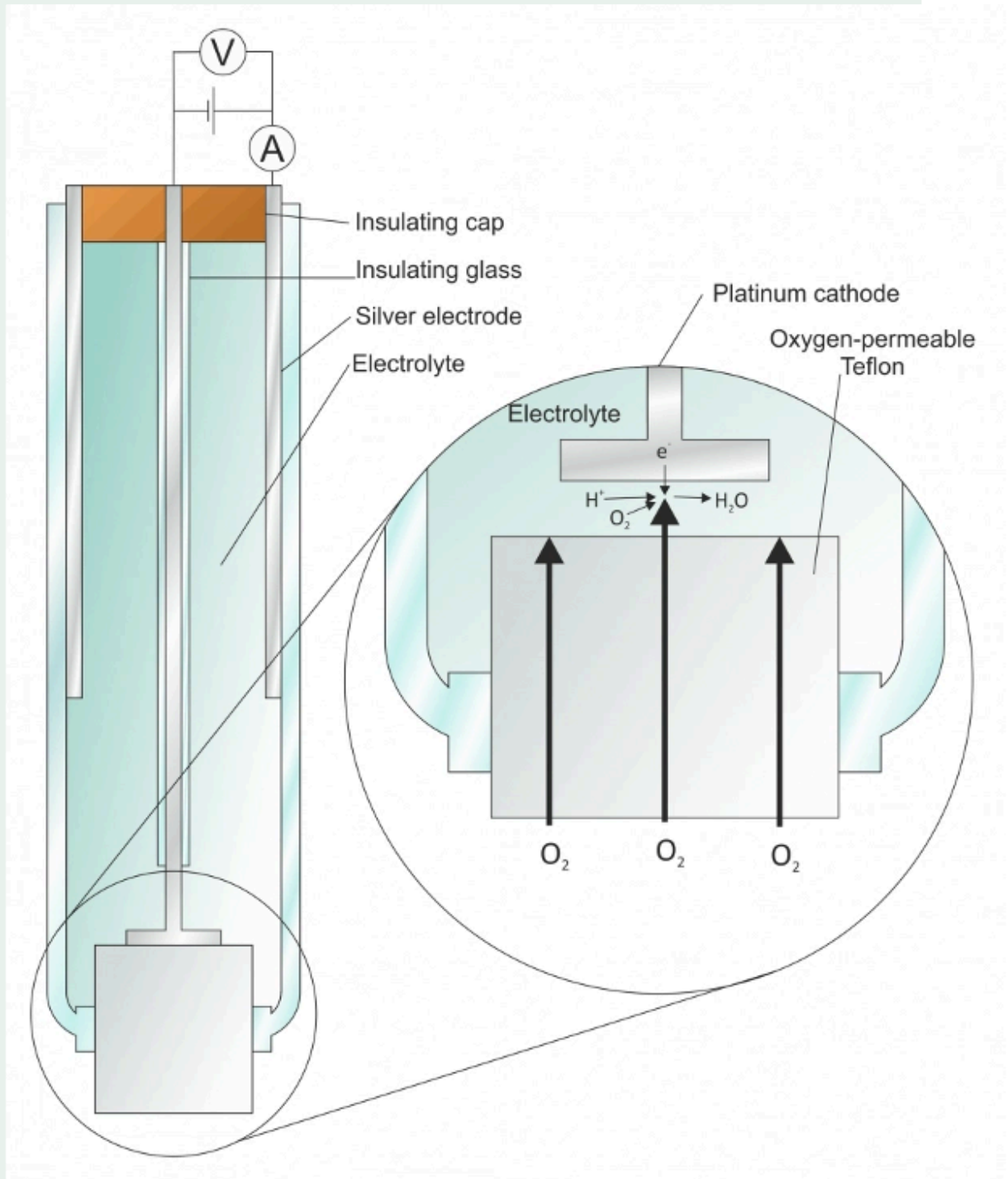
- It depends on the **direction of proton flow across the inner mitochondrial membrane**
- Protons flowing from space to matrix will trigger the making of ATP
- Protons flowing from matrix to space will trigger the consumption of ATP.

▼ What is the purpose of the oxygen electrode?

- To **measure the concentration of oxygen within a solution**
 - This hence allows us to **measure oxidative phosphorylation by monitoring the consumption of oxygen**

▼ Which metals are present at the cathode and anode?

- Cathode - platinum
- Anode - silver



▼ What happens at the platinum cathode?

- $O_2 + 4H^+ + 4e^- \rightarrow 2 H_2O$
- **Oxygen is reduced to water**

▼ What happens at the silver anode?

- $4 Ag_0 + 4Cl^- \rightarrow AgCl + 4e^-$
- This is **to complete the circuit**, so there is current running through it

▼ How does the oxygen electrode allow us to measure the rate of oxidative phosphorylation?

- It is a device that allows us to measure oxygen levels in a solution
- By measuring the rate of oxygen consumption in mitochondria, we can measure the rate of oxidative phosphorylation

▼ What does respiratory control describe in oxidative phosphorylation?

- It describes how **ADP and inorganic phosphate** control the rate of oxygen consumption in oxidative phosphorylation
- This allows the body to **adjust the rate of oxygen consumption** according to **how much energy is required by the body**

▼ How is oxygen then reduced to water in oxidative phosphorylation?

- The e^- from the electron transport chain and the H^+ that move across the inner mitochondrial membrane are added to O_2 to form water

▼ What is a metabolic poison?

- A molecule that interrupts the movement of e^- across the electron transport chain, or interrupts the movement of H^+ across the inner membrane to generate ATP
- Both methods stop ATP from being synthesised which is their purpose.

▼ How do cyanide (CN^-) and azide act as metabolic poisons?

- It binds to the **haem group in complex IV** of the electron transport chain which prevents electrons from flowing through
- This ultimately stops ATP from being formed.

▼ How does malonate act as a metabolic poison?

- Malonate **slows down the flow of electrons** from **complex II (succinate dehydrogenase)** to **coenzyme Q (ubiquinone)**
- It does this by **inhibiting the oxidation of succinate to fumarate (step 6 of TCA Cycle)**.

▼ How can dinitrophenol (DNP) be used to induce weight loss?

- DNP works by pumping protons across the inner mitochondrial membrane and ensuring that **the energy produced from this movement is not used by ATP synthase to make ATP**
- This then means that a lot of protons pumped are actually wasted (not used to make ATP), and more protons hence need to be pumped than usual to make the same amount of ATP
- This means more electrons need to be pumped across the electron transport chain, more cofactors need to be reoxidised and hence produced and so **more fuel (glucose, fatty acids) needs to be metabolised to produce similar amounts of ATP**

▼ Why is DNP no longer prescribed?

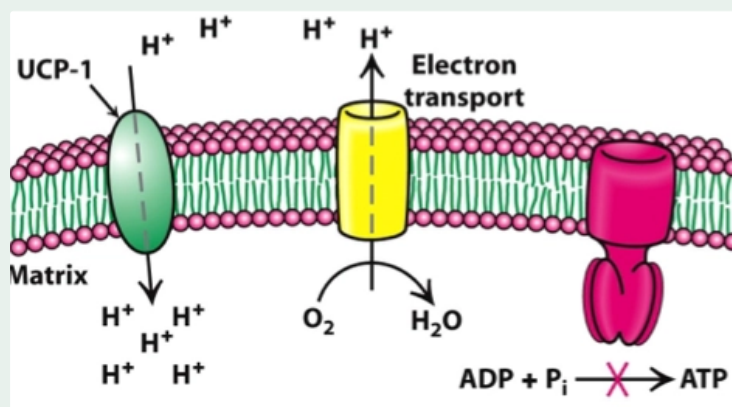
- The difference between the dose which is considered effective and that which is considered fatal is so slight
- Many patients ended up dying or suffering fatal injury before it was banned in 1937

▼ What is non-shivering thermogenesis?

- It is a process which (like DNP) uncouples ATP from proton pumping in response to low body temperatures

▼ How does non-shivering thermogenesis work?

- An alternative proton channel called **UCP-1** is **activated in response to low temperatures**
- Protons are pumped through this channel following an electrochemical gradient and instead of the **energy from this movement** being used to form ATP via ATP synthase, it is **dissipated as heat energy**



▼ What is rotenone and how does it work?

- It is also a metabolic poison which works by **preventing electron transfer from complex I to coenzyme Q (ubiquinone)**

▼ What is oligomycin and how does it work?

- It **binds to the stalk of ATP synthase** and **prevents H⁺ flowing through it**
- This obviously **stops ATP synthesis** via the enzyme

▼ What is cytochrome C?

- Cytochrome C is an **electron transport chain protein** that **transfers electrons between complex III and complex IV** of the chain.

